NVIDIA NCP-AII

AI Infrastructure 专业认证

75题中英双语原创模拟题库
75-Question Chinese-English Practice Bank

版本	2026-08-21
考试模拟	75题 / 建议120分钟
内容	题目、作答栏、答案与逐题解析
声明	原创练习资料，非NVIDIA官方试题或考试泄题

使用说明 / Instructions

- 先完成第一部分的75题，再翻到第二部分核对答案。
- 单选题只有一个最佳答案；多选题必须选出全部正确选项才计分。
- 建议限时120分钟。每题1分：60分以下需重建基础，60–67分接近，68分以上表现良好。
- 考试与费用可能更新，报名时请以NVIDIA官方页面为准。

领域分布 / Domain Distribution

考试领域	模拟题数	官方权重
系统与服务器启动	23	31%
实体层管理	4	5%
控制平面安装与配置	14	19%
集群测试与验证	25	33%
故障排除与优化	9	12%

第一部分：模拟试题 / Part I: Practice Questions

第一领域：系统与服务器启动（23题）
Domain 1: System and Server Bring-up (23 questions)

1. [单选] 新安装GPU服务器后，要快速确认驱动、GPU型号、显存和运行状态，应先执行哪一个命令?

1. [Single choice] Which command should be executed after the installation of the new GPU server, in order to quickly confirm the driver, the GPU model, GPU memory and the operational status?

A. lscpu

A. lscpu

B. nvidia-smi

B. nvidia-smi

C. nvcc --build

C. nvcc --build

D. systemctl status docker

D. systemctl status docker

作答 / Answer：_____

2. [单选] BMC在GPU服务器部署中的主要用途是什么?

2. [Single choice] What are the main uses of BMC in GPU server deployment?

A. 编译CUDA程序

A. Compile CUDA Program

B. 在主机操作系统之外进行远程监控与管理

B. Remote monitoring and management outside the mainframe operating system

C. 执行NCCL集合通信

C. Execute NCCL assembly communications

D. 创建容器镜像

D. Create container mirror

作答 / Answer：_____

3. [单选] OOB管理网络的主要价值是什么?

3. [Single choice] What are the main values of the OOB management network?

A. 让训练流量获得更高带宽

A. Make the training stream more bandwidth.

B. 在数据网络或操作系统异常时仍可管理设备

B. Equipment to manage data networks or operating systems in case of anomalies

C. 取代所有交换网络

C. Replace all exchange networks

D. 自动分割GPU

D. Auto Split GPU

作答 / Answer：_____

4. [单选] TPM最直接用于支持哪项能力?

4. [Single choice] What capability is the TPM most directly used to support?

A. 硬件信任根和安全启动相关密钥保护

A. Hardware Trust Roots and Security Start Related Key Protection

B. GPU散热控制

B. GPU dissipation control

C. InfiniBand路由

C. InfiniBand route

D. 存储条带化

D. Storage Belging

作答 / Answer：_____

5. [多选] 服务器上架通电前，哪些检查合理?

5. [Multiple choice] Which checks are reasonable before the server is installed?

A. 确认机柜承重

A. Make sure the cabinet weighs.

B. 核对电源回路容量

B. Check the power circuit capacity.

C. 确认进出风方向

C. Make sure you get in and out.

D. 忽略设备接地

D. Ignore device location

作答 / Answer：_____

6. [单选] 更新HGX平台固件前，最重要的准备动作是什么?

6. [Single choice] What are the most important preparatory actions before updating the HGX platform firmware?

A. 直接强制断电

A. Directly forced power outage

B. 核对兼容性矩阵、版本与回退方案

B. Check Compatibility Matrix, Versions and Return Schemes

C. 删除BMC日志

C. Delete BMC Log

D. 关闭所有风扇

D. Close All Fans

作答 / Answer：_____

7. [单选] GPU服务器功耗持续接近机柜回路上限，最适合先做什么？

7. [Single choice] The utility of the GPU server continues to be close to the upper limit of the locker and is best suited for what to do first?

A. 增加训练批次

A. Increase in training batches

B. 验证电源预算、冗余模式及实际峰值

B. Validation of power budget, redundancy patterns and actual peaks

C. 禁用所有监控

C. Disable all monitors

D. 移除接地线

D. Remove Ground Line

作答 / Answer：_____

8. [单选] 热通道/冷通道规划的主要目的是什么？

8. [Single choice] What is the main purpose of the heat/cold route planning?

A. 混合冷热空气

A. Combining cold and hot air

B. 减少冷热气流混合并提高冷却效率

B. Reduce cold and hot air stream mixing and increase cooling efficiency

C. 提升磁盘容量

C. Raise Disk Capacity

D. 替代液冷系统

D. Alternative liquid cooling system

作答 / Answer：_____

9. [多选] 验证新装GPU硬件时应检查哪些项目？

9. [Multiple choice] Which items should be checked when testing new GPU hardware?

A. PCIe枚举

A. PCIE Enumeration

B. GPU温度和功耗

B. GPU temperature and power consumption

C. ECC错误

C. ECC Error

D. NVLink状态

D. NVLink Status

作答 / Answer：_____

10. [单选] lspci命令在GPU安装验证中主要用来做什么?

10. [Single choice] What was the lspci command used primarily in the GPU installation verification?

A. 查看PCIe设备是否被系统枚举

A. See if the PCIe equipment is charged to the system

B. 测试存储吞吐量

B. Test storage throughput

C. 更新GPU固件

C. Update GPU solid

D. 配置Slurm队列

D. Configure Slurum Queue

作答 / Answer：_____

11. [单选] 发现一张GPU未出现在lspci输出中，最可能优先检查哪一层?

11. [Single choice] It was found that a GPU did not appear in the lspci output, which level was the most likely priority to check?

A. 应用程序参数

A. Application parameters

B. 物理安装、供电、插槽与BIOS资源配置

B. Physical installation, power supply, slots and BIOS resource allocation

C. Python版本

C. Python version

D. NGC账号

D. NGC Account

作答 / Answer：_____

12. [单选] PCIe链路降速时，哪个信息最有助于确认协商宽度与速率?

12. [Single choice] Which information is most helpful in confirming the width and speed of the consultation when the PCIe link is down?

A. lspci -vv

A. lspci -vv

B. pwd

B. pwd

C. docker ps

C. docker ps

D. ip route

D. ip route

作答 / Answer：_____

13. [多选] 选择高速网络收发器时必须核对哪些条件?

13. [Multiple choice] What conditions must be checked when choosing a high-speed network transceiver?

A. 速率与协议

A. Rates and protocols

B. 光纤类型和波长

B. Fibre-optic type and wavelength

C. 连接器与线缆长度

C. Connector and Wire Length

D. 交换机和网卡支持矩阵

D. Switches and webcard support matrix

作答 / Answer：_____

14. [单选] 将单模光模块配到不适用的多模光纤，最可能造成什么?

14. [Single choice] What is most likely to happen if single-modular modules are matched to non-applicable multi-modular fibre-optics?

A. 自动提升带宽

A. Automatically raise bandwidth

B. 链路不起或信号质量异常

B. Can't connect or signal quality is abnormal.

C. GPU显存增加

C. GPU display increase

D. BMC自动升级

D. BMC Auto Upgrade

作答 / Answer：_____

15. [单选] AI Factory常采用叶脊（leaf-spine）拓扑，主要原因是什么?

15. [Single choice] AI Factory often uses leaf-spine, the main reason for which?

A. 只允许南北向流量

A. Only North-South traffic is allowed.

B. 提供可预测的东西向带宽与扩展性

B. Provide predictable East-West bandwidth and extension

C. 不需要路由

C. No route required.

D. 可省略所有交换机

D. Save All Switches

作答 / Answer：_____

16. [单选] 计算网络与管理网络分离的主要好处是什么?

16. [Single choice] What are the main benefits of separating the network from the management network?

A. 让所有流量共享故障域

A. Let all traffic share faulty fields

B. 隔离业务与管理流量并降低相互影响

B. Segregation of operations and management of traffic and reduction of interaction

C. 删除访问控制

C. Remove Access Control

D. 取消IP规划

D. Cancel IP planning

作答 / Answer：_____

17. [多选] 初始BMC配置应包含哪些内容?

17. [Multiple choice] What should be included in the initial BMC configuration?

A. 设置唯一管理IP

A. Set Unique Manage IP

B. 修改默认凭证

B. Modify Default Certificate

C. 配置时间和日志

C. Configure Time and Logs

D. 限制管理访问来源

D. Limiting access to sources

作答 / Answer：_____

18. [单选] 服务器事件日志出现重复电源供应器掉线，首先应如何处理?

18. [Single choice] What should be done in the first place when the server event log is running a duplicate power supply out of line?

A. 删除日志后忽略

A. Ignore after deleting the log

B. 检查PSU状态、电源线、PDU端口与冗余供电

B. Check PSU status, power lines, PDU ports and redundant power supply

C. 重装Python

C. Reload Python

D. 修改容器标签

D. Modify container labels

作答 / Answer：_____

19. [单选] 安装双宽GPU时，除了插槽位置还必须特别确认什么？

19. [Single choice] When installing double width GPUs, what must be specifically identified in addition to the slot position?

A. 辅助供电、固定结构和散热间距

A. Supplementary power supply, fixed structure and heat-dispersion spacing

B. 浏览器版本

B. Browser Version

C. DNS搜索域

C. DNS Search Field

D. 用户shell

D. User shell

作答 / Answer：_____

20. [单选] 第三方存储初始接入GPU集群时，最先应确认什么？

20. [Single choice] What should be identified first when initial access to GPU clusters is stored by third parties?

A. 容量名称是否好看

A. Does the capacity name look good?

B. 网络、认证、挂载参数和支持的文件系统

B. Network, authentication, mounted parameters and supporting documentation systems

C. GPU风扇颜色

C. GPU fan colour

D. BMC背景图

D. BMC background

作答 / Answer：_____

21. [多选] 服务器交付验收记录应包含哪些内容？

21. [Multiple choice] What should be included in the receipt and inspection records of the server delivery?

A. 序号与资产信息

A. Serial number and asset information

B. 固件和驱动版本

B. Solidware and Driver Version

C. 验证结果与异常

C. Authentication results and anomalies

D. 线缆和端口映射

D. Wire and port map

作答 / Answer：_____

22. [单选] 固件升级后设备异常，最合理的下一步是什么？

22. [Single choice] What is the most logical next step after the upgrade of the solidware?

A. 继续随机升级

A. Continue random upgrades

B. 查看事件日志并与支持矩阵比对，必要时按计划回退

B. View the event log and compare it with the support matrix and return as planned if necessary

C. 删除全部配置

C. Remove All Configurations

D. 关闭监控

D. Turn off the surveillance.

作答 / Answer：_____

23. [单选] 完成服务器bring-up后，进入集群部署前最重要的结果是什么？

23. [Single choice] What are the most important results before entering the cluster before completion of the bling-up server?

A. 有可重复的硬件、固件、网络与健康验证基线

A. Repetitible hardware, solidware, network and health validation baselines

B. 只有主机名

B. Only hostname

C. 只有一张照片

C. There's only one picture.

D. 只确认电源灯亮

D. Just make sure the power lights are on.

作答 / Answer：_____

第二领域：实体层管理（4题）
Domain 2: Physical Layer Management (4 questions)

24. [单选] MIG的主要功能是什么？

24. [Single choice] What's MiG's main function?

A. 合并多台服务器

A. Merge multiple servers

B. 将支持的GPU划分为具有硬件隔离的实例

B. Split supported GPUs into hardware-segregated examples

C. 更新交换机固件

C. Update switch solids

D. 压缩模型文件

D. Compression Model File

作答 / Answer：_____

25. [多选] 规划MIG时应考虑哪些因素?

25. [Multiple choice] What factors should be taken into account in the planning of the MIG?

A. 工作负载所需计算和显存

A. Calculation and display of task loads

B. 支持的GPU型号与profile

B. Supported GPU Models and Profile

C. 调度器资源发现

C. Scheduler resource found

D. 实例隔离与监控需求

D. Examples of isolation and surveillance needs

作答 / Answer：_____

26. [单选] BlueField DPU的典型价值是什么?

26. [Single choice] What is the typical value of BlueField DPU?

A. 把网络、存储与安全服务从主机CPU卸载

A. Unmount the network, storage and security services from the host CPU

B. 替代所有GPU

B. Replace all GPUs

C. 增加CPU核心频率

C. Increase CPU core frequency

D. 只用作显示卡

D. Use only as a display card

作答 / Answer：_____

27. [单选] 变更MIG配置后，调度器仍显示旧资源，最应该做什么?

27. [Single choice] With the change in MIG configuration, the scheduler still shows old resources, and what should be done most?

A. 忽略差异

A. Ignore differences

B. 更新或重启相关GPU资源发现组件并验证资源注册

B. Update or restart relevant GPU resource discovery components and validate resource registration

C. 删除BMC

C. Delete BMC

D. 降低风扇转速

D. Lower fan turn.

作答 / Answer：_____

第三领域：控制平面安装与配置（14题）
Domain 3: Control Plane Installation and Configuration (14 questions)

28. [单选] BCM在AI集群中的主要作用是什么?

28. [Single choice] What is the main role of the BCM in the AI cluster?

A. 集中进行节点部署、配置与集群生命周期管理

A. Focused nodal deployment, configuration and cluster life cycle management

B. 只编辑图片

B. Edit Pictures Only

C. 替代GPU驱动

C. Replace GPU Drive

D. 只提供DNS查询

D. Only DNS queries

作答 / Answer：_____

29. [单选] BCM高可用配置最重要的目标是什么?

29. [Single choice] What are the most important objectives of the high-availability configuration of BCM?

A. 消除管理服务单点故障

A. Elimination of single point failure of management services

B. 增加网页颜色

B. Add page colours

C. 禁用数据库同步

C. Disable Database Sync

D. 让所有节点共用一个电源

D. Let all nodes share a power source.

作答 / Answer：_____

30. [多选] 操作系统批量安装前应准备哪些内容?

30. [Multiple choice] What should be prepared before bulk installation of the operating system?

A. 支持的OS镜像

A. Supported OS mirrors

B. 网络启动/DHCP规划

B. Network start-up/DHCP planning

C. 节点角色与分区

C. Node Roles and Divisions

D. 驱动和固件兼容性

D. Driving and solidware compatibility

作答 / Answer：_____

31. [单选] Slurm中sinfo主要用于查看什么?

31. [Single choice] What does the slurm sinfo use to see?

A. 分区和节点状态摘要

A. Summary status of partitions and nodes

B. GPU固件内容

B. GPU Solidware Contents

C. 光功率

C. Photopower

D. Docker镜像层

D. Docker Mirror Layer

作答 / Answer：_____

32. [单选] Slurm中squeue主要用于查看什么?

32. [Single choice] What's the main slurm squeue used to see?

A. 当前排队和运行中的作业

A. Current queuing and running jobs

B. BIOS设置

B. BIOS Settings

C. BMC传感器

C. BMC Sensor

D. PCIe链路能力

D. PCIe Linking Capacity

作答 / Answer：_____

33. [单选] Enroot与Pyxis组合的典型用途是什么?

33. [Single choice] What are the typical uses of Enroot and Pyxis?

A. 在Slurm作业中运行容器化工作负载

A. Run containerization loads in the Slurm operation

B. 替代机柜PDU

B. Alternative PDU

C. 配置TPM

C. Configure TPM

D. 测量光功率

D. Measuring photopower

作答 / Answer：_____

34. [单选] 安装NVIDIA驱动后nvidia-smi无法通信，且刚升级内核，最可能是什么?

34. [Single choice] What is most likely the installation of the NVIDIA-driven nvidia-smi, which cannot communicate and which has just been upgraded to the kernel?

A. 内核模块未为当前内核正确构建或加载

A. kernel module is not properly constructed or loaded for the current kernel

B. DNS故障

B. DNS failure.

C. 用户目录太大

C. The user directory is too big

D. NGC密码过短

D. NGC codes are too short.

作答 / Answer：_____

35. [单选] DOCA主要面向哪类平台能力?

35. [Single choice] What kind of platform capacity does DOCA focus on?

A. BlueField DPU上的数据中心基础设施加速

A. Accelerated data centre infrastructure on BlueField DPU

B. 文字处理

B. Text Processing

C. CPU超频

C. CPU Overfrequency

D. 显示器校色

D. Display Colour

作答 / Answer：_____

36. [单选] 安装NVIDIA Container Toolkit后，验证Docker使用GPU的正确方式是哪一个?

36. [Single choice] What's the correct way to verify Docker's use of GPU after the installation of NVIDIA Container Toolkit?

A. docker run --rm --gpus all nvidia/cuda:<tag> nvidia-smi

A. docker run --rm --gpus all nvidia/cuda:<tag> nvidia-smi

B. docker gpu enable

B. docker gpu enable

C. nvidia-smi --docker

C. nvidia-smi --docker

D. docker run --cpu gpu

D. docker run --cpu gpu

作答 / Answer：_____

37. [多选] 使用NGC CLI前通常需要哪些准备?

37. [Multiple choice] What are the usual preparations for use of NGC CLI?

A. 安装并初始化NGC CLI

A. Install and Initialize NGC CLI

B. 配置API Key

B. Configure API Key

C. 确认网络可访问所需服务

C. Confirm that the network has access to services required

D. 核对组织或团队权限

D. Check organizational or team privileges

作答 / Answer：_____

38. [单选] Kubernetes节点要让Pod使用NVIDIA GPU，关键组件是什么?

38. [Single choice] What are the key components of the Kubernetes node for Pod to use NVIDIA GPU?

A. NVIDIA GPU Operator或相应device plugin及驱动栈

A. NVIDIA GPU Operation or equivalent Device plugin and drive

B. Apache网页服务器

B. Apache web server

C. Samba客户端

C. Samba Client

D. 打印服务

D. Printing services

作答 / Answer：_____

39. [单选] 容器内CUDA版本与主机驱动的关系，正确的是哪一项?

39. [Single choice] Which CUDA version of the container is correct in relation to the mainframe drive?

A. 主机驱动必须满足容器CUDA所需的最低兼容要求

A. Host driver must meet minimum compatibility requirements for container CUDA

B. 两者完全无关

B. It has nothing to do with it.

C. 容器会自动替换主机内核驱动

C. The container automatically replaces the nuclear drive in the host.

D. 只看Python版本

D. Just the Python version.

作答 / Answer：_____

40. [多选] 集群节点分类（category）通常适合定义什么？

40. [Multiple choice] What would cluster nodes (category) normally be appropriate for definition?

A. 角色一致的镜像

A. The mirror of the same character.

B. 网络接口配置

B. Network Interface Configuration

C. 软件包与配置覆盖

C. Package & Configure Overwrite

D. 节点组的共同属性

D. Common Properties for Node Groups

作答 / Answer：_____

41. [单选] 控制平面安装完成的最佳验证方式是什么？

41. [Single choice] What is the best way to control the completion of the plane installation?

A. 从供应、节点注册、调度到容器GPU作业执行进行端到端测试

A. Conduct end-to-end testing from supply, nodal registration, dispatch to container GPU operations

B. 只看登录页面

B. Read Login Pages Only

C. 只ping localhost

C. Only ping localhost

D. 只确认硬盘容量

D. Only confirm hard disk capacity

作答 / Answer：_____

第四领域：集群测试与验证（25题）
Domain 4: Cluster Test and Verification (25 questions)

42. [单选] HPL主要测量哪项能力？

42. [Single choice] What capacity does HPL measure primarily?

A. 稠密线性代数的FP64计算性能

A. FP64 computational performance for dense linear algebras

B. DNS延迟

B. DNS delay

C. 网页加载速度

C. Web Load Speed

D. 磁盘权限

D. Disk Permissions

作答 / Answer：_____

43. [单选] HPL结果中的Rmax表示什么?

43. [Single choice] What does Rmax mean in HPL results?

A. 测试实际达到的最高性能

A. Test the best performance actually achieved

B. 理论峰值功耗

B. Theoretical peak effort

C. 网络最大跳数

C. Maximum number of network jumps

D. GPU数量

D. Number of GPUs

作答 / Answer：_____

44. [单选] HPL burn-in除性能外还可帮助发现什么?

44. [Single choice] What else can HPL burn-in help to discover besides performance?

A. 满载下的散热、供电及稳定性问题

A. Full load of heat, electricity and stability issues

B. 用户拼写错误

B. User spelling error

C. 浏览器书签

C. Browser Bookmarks

D. 邮件配置

D. Mail Configuration

作答 / Answer：_____

45. [单选] NCCL Tests主要验证什么?

45. [Single choice] What does NCCL Tests do?

A. GPU间集合通信带宽与延迟

A. Cluster bandwidth and delay between GPUs

B. CPU文件压缩

B. CPU File Compression

C. BMC密码复杂度

C. BMC password complexity

D. RAID容量

D. RAID Capacity

作答 / Answer：_____

46. [单选] 多节点NCCL all-reduce性能异常低，优先检查什么？

46. [Single choice] What's the priority for multinode NCCL all-reduce performance?

A. GPU拓扑、网络接口选择、链路状态和进程映射

A. GPU scale-up, web interface selection, chain status and process mapping

B. 浏览器缓存

B. Browser Cache

C. 主机壁纸

C. Host Wallpaper

D. 文本编辑器

D. Text Editor

作答 / Answer：_____

47. [单选] nvidia-smi topo -m最适合用来查看什么？

47. [Single choice] What's best for a nvidia-smi topo-m?

A. GPU、NIC、CPU之间的拓扑关系

A. The relationship between GPU, NC, CPU.

B. 作业队列

B. Job Queue

C. 文件系统容量

C. File System Capacity

D. BMC用户

D. BMC users

作答 / Answer：_____

48. [多选] 单节点压力测试期间应监控哪些指标？

48. [Multiple choice] What indicators should be monitored during single-point stress tests?

A. 温度与功耗

A. Temperature and power consumption

B. 时钟和降频原因

B. Clock and downtime causes

C. ECC/Xid错误

C. ECC/Xid error

D. GPU利用率

D. GPU utilization

作答 / Answer：_____

49. [单选] DCGM诊断的主要价值是什么?

49. [Single choice] What is the main value of DCGM diagnosis?

A. 对GPU健康与关键子系统进行标准化诊断

A. Standardized diagnostic of GPU health and critical subsystems

B. 配置网页DNS

B. Configure web page DNS

C. 编译Linux内核

C. Compile Linux kernel

D. 替代所有应用基准

D. Replace all applied benchmarks

作答 / Answer：_____

50. [单选] ClusterKit通常用于什么?

50. [Single choice] What does ClusterKit usually use?

A. 对集群节点执行多方面评估与验证

A. Multi-faceted assessment and validation of cluster nodes

B. 创建用户头像

B. Create user header

C. 修改交换机外壳

C. Modify switch shell

D. 压缩日志

D. Compressed Log

作答 / Answer：_____

51. [多选] 验证InfiniBand链路时哪些资料有用?

51. [Multiple choice] What is useful to verify the Infiniband link?

A. 端口状态与速率

A. Port status and rate

B. 错误计数器

B. Error counter

C. 线缆/收发器信息

C. Wire/receiver information

D. 子网管理器状态

D. Subnet Manager Status

作答 / Answer：_____

52. [单选] ibstat通常用于查看什么?

52. [Single choice] What does ibstat normally use to see?

A. InfiniBand HCA和端口状态

A. InfiniBand HCA and Port Status

B. GPU显存

B. GPU memory

C. Docker层

C. Docker floor.

D. BIOS密码

D. BIOS password

作答 / Answer：_____

53. [单选] 交换机端口持续产生CRC错误，最合理的优先动作是什么?

53. [Single choice] What is the most logical priority action for the exchange port, which continues to cause CRC errors?

A. 检查并重插或替换线缆/模块，同时确认端口信号质量

A. Check and insert or replace cables/modules while confirming port signal quality

B. 重装Python

B. Reload Python

C. 增加训练epochs

C. Increased training epochs

D. 删除用户

D. Remove User

作答 / Answer：_____

54. [单选] 为什么要核对交换机固件与软件版本?

54. [Single choice] Why check the exchange solids with the software version?

A. 确保功能、修复与互操作性符合支持矩阵

A. Ensuring functionality, repair and interoperability in line with the support matrix

B. 增加GPU显存

B. Increase GPU Reveal

C. 改变机柜高度

C. Change the height of the cabinet

D. 替代布线记录

D. Alternative Wire Record

作答 / Answer：_____

55. [单选] 验证BlueField-3时除了固件还应确认什么?

55. [Single choice] What should BlueField-3 be able to confirm in addition to solids?

A. 工作模式、驱动、端口与主机/DPU连通状态

A. Work mode, driver, port-to-host/DPU connectivity

B. 文档页码

B. Document Page Number

C. 浏览器主题

C. Browser Theme

D. 键盘布局

D. Keyboard Layout

作答 / Answer：_____

56. [多选] 收发器固件与健康验证可能包含哪些内容?

56. [Multiple choice] What might be the content of transceivers solids and health certification?

A. 固件版本

A. Solidware Version

B. 温度与光功率

B. Temperature and light power

C. 错误告警

C. Wrong alarm.

D. 设备支持状态

D. Device support status

作答 / Answer：_____

57. [单选] E/W fabric在AI集群中通常指什么?

57. [Single choice] What does E/W fabric usually mean in the AI cluster?

A. 节点之间的东西向高性能网络

A. Something between nodes leads to a high-performance network.

B. 外部互联网出口

B. External Internet exports

C. BMC电源线

C. BMC power lines

D. 显示输出

D. Show Output

作答 / Answer：_____

58. [单选] NCCL burn-in运行一段时间后出现单节点掉速，下一步应做什么?

58. [Single choice] What should be done next when the NCCL burn-in is running at a single node after a certain period of time?

A. 对比该节点拓扑、温度、错误日志和网络计数器

A. Compare the node scale, temperature, error log and network counter

B. 忽略并发布生产

B. Ignore and release production

C. 删除全部结果

C. Remove All Results

D. 只重命名节点

D. Rename Node Only

作答 / Answer：_____

59. [单选] NeMo burn-in相较HPL更接近哪类验证?

59. [Single choice] What kind of authentication is NeMo burn-in closer to HPL?

A. AI模型工作负载与软件栈的综合稳定性

A. Consolidated stability of AI work load and software stacks

B. 纯DNS服务

B. DNS service

C. 机柜承重

C. Loading machine cabinets

D. BMC认证

D. BMC Certification

作答 / Answer：_____

60. [多选] 存储性能测试应关注哪些指标?

60. [Multiple choice] What indicators should be the focus of storage performance testing?

A. 吞吐量

A. Swallow Volume

B. IOPS

B. IOPS

C. 延迟

C. Delay

D. 多客户端扩展性

D. Multi-client extension

作答 / Answer：_____

61. [单选] 单客户端存储很快，但多节点训练变慢，最需要测试什么？

61. [Single choice] Single-client storage is fast, but multi-node training slows, and what is most needed to test?

A. 多客户端并发吞吐及元数据服务能力

A. Multi-Customs and Spam and Metadata Services Capability

B. 屏幕分辨率

B. Screen Resolution

C. BMC主题

C. BMC Theme

D. CPU主机名

D. CPU hostname

作答 / Answer：_____

62. [单选] 测试结果可重复性的关键条件是什么？

62. [Single choice] What are the key conditions for replicability of test results?

A. 固定版本、参数、拓扑和环境并保存原始结果

A. Fixed version, parameters, pouncer and environment and preservation of original results

B. 每次随机换参数

B. Randomly change parameters at each time

C. 不记录日志

C. Do Not Record Log

D. 只记最后数字

D. Only the last number.

作答 / Answer：_____

63. [多选] 一次完整集群验收应涵盖哪些层次？

63. [Multiple choice] What levels should be covered by a full cluster inspection?

A. 单节点健康

A. Single Node Health

B. GPU内部/节点内通信

B. GPU internal/inline communications

C. 跨节点网络

C. Cross-Node Network

D. 存储与代表性AI负载

D. Storage and Representative AI Load

作答 / Answer：_____

64. [单选] NCCL带宽符合预期但训练速度仍低，接下来最合理检查什么？

64. [Single choice] What is the most reasonable next check for NCCL bandwidth that meets expectations but is still training at a low rate?

A. 数据输入、CPU/NUMA亲和、框架配置和计算利用率

A. Data entry, CPU/NUMA parentage, framework configuration and calculation utilization

B. 只换网线

B. Just switch lines.

C. 删除BMC

C. Delete BMC

D. 关闭日志

D. Close Log

作答 / Answer：_____

65. [单选] 验证线缆连接正确性的最佳方法是什么？

65. [Single choice] What is the best way to verify the correctness of cable connections?

A. 将设计端口映射与实际LLDP/链路信息及标签逐一比对

A. Match the design port map with actual LLDP/link information and labels

B. 仅观察灯是否亮

B. Just watch if the lights are on.

C. 猜测连接

C. Guess Connection

D. 只检查一端

D. Check only one end

作答 / Answer：_____

66. [单选] 集群验证通过的合格标准应如何设定？

66. [Single choice] How should the eligibility criteria adopted for cluster validation be set?

A. 在测试前依据设计规格定义明确阈值和允许偏差

A. Define clear thresholds and permitted deviations based on design specifications before testing

B. 测完再随意决定

B. We'll make up our mind when we're done.

C. 只要命令退出即可

C. Just give the order to quit.

D. 只比较GPU数量

D. Compare GPU numbers only

作答 / Answer：_____

第五领域：故障排除与优化（9题）
Domain 5: Troubleshoot and Optimize (9 questions)

67. [单选] nvidia-smi显示一张GPU丢失，lspci也看不到，优先检查什么?

67. [Single choice] nvidia-smi shows a GPU missing and lspci can't see it. What's the priority?

A. 物理连接、供电、插槽和平台日志

A. Physical connection, power supply, slots and platform logs

B. Python虚拟环境

B. Python Virtual Environment

C. DNS缓存

C. DNS Cache

D. NGC组织名

D. NAMC Organization Name

作答 / Answer：_____

68. [单选] GPU出现Xid错误时，最佳第一步是什么?

68. [Single choice] What's the best first step when GPU has a Xid error?

A. 记录Xid代码、时间与工作负载并查询相关日志

A. Record Xid codes, time and task loads and query related logs

B. 立即删除所有日志

B. Remove all logs immediately

C. 永久忽略

C. Ignore permanently

D. 修改主机名

D. Modify hostname

作答 / Answer：_____

69. [多选] 判断GPU是否发生热降频可检查哪些资料?

69. [Multiple choice] What information can GPU be examined for determining whether thermal precipitation is occurring?

A. 温度

A. Temperature

B. 时钟频率

B. Clock Frequency

C. throttle reasons

C. throttle reasons

D. 功耗限制

D. Work-use limit

作答 / Answer：_____

70. [单选] 某节点NCCL性能低且NIC位于远端NUMA节点，适合采取什么优化?

70. [Single choice] What could be done to optimize the low performance of NCCL at a node and NIC at a remote NUMA node?

A. 调整进程、CPU、GPU和NIC亲和关系

A. Adjustment process, CPUs, GPUs and NCs

B. 关闭所有CPU核心

B. Close all CPU cores

C. 删除路由表

C. Remove Route Table

D. 降低链路速率

D. Lower the chain rate.

作答 / Answer：_____

71. [单选] 存储吞吐不足但网络仍有余量，下一步最适合检查什么?

71. [Single choice] What's the next step best to check if there's not enough storage but there's still more to the network?

A. 文件系统参数、并发度、条带配置和后端磁盘瓶颈

A. File system parameters, conjunction, strip configuration and back disk bottlenecks

B. GPU颜色

B. GPU colour

C. BMC语言

C. BMC language

D. SSH横幅

D. SSH banner

作答 / Answer：_____

72. [单选] 发现冗余PSU之一故障，正确处理方式是什么?

72. [Single choice] What's the correct way of dealing with one of the redundant PSU malfunctions?

A. 按维护流程确认替换件并在保持冗余风险可控下更换

A. Identification of replacements by maintenance process and controlled replacements to maintain risk of redundancy

B. 拔掉另一颗PSU

B. Pull out the other PSU.

C. 忽略到全部断电

C. Ignore all power outages

D. 用软件更新修复风扇

D. Refurbish fans with software update

作答 / Answer：_____

73. [多选] 替换故障GPU后应完成哪些验证?

73. [Multiple choice] What certification should be completed after replacing the malfunctioning GPU?

A. PCIe枚举和nvidia-smi

A. PCIe & nvidia-smi

B. 固件/驱动兼容性

B. Solidware/Driving Compatibility

C. DCGM或压力测试

C. DCGM or pressure test

D. NVLink/NCCL相关测试

D. NVLink/NCCL-related testing

作答 / Answer：_____

74. [单选] Intel或AMD双路服务器优化时，最关键的通用原则是什么?

74. [Single choice] What are the most critical common principles when intel or AMD dual-circuit servers are optimized?

A. 了解NUMA拓扑并让CPU、内存、GPU、NIC与进程保持合理亲和

A. To understand the NUMA and keep the CPU, memory, GPU, NC and process reasonably close.

B. 随机绑定核心

B. Randomly bind core

C. 永久关闭监控

C. Permanently shut down surveillance.

D. 所有作业只用一个核心

D. All the work is in one core.

作答 / Answer：_____

75. [单选] 排错完成后，哪项做法最能减少问题再次发生?

75. [Single choice] What would be the best way to reduce the recurrence of the problem once the errors are completed?

A. 记录根因、修复、验证证据并更新基线与自动检查

A. Recording of root causes, repairing, validation of evidence and updating of baselines and automated checks

B. 删除工单

B. Remove Work Sheet

C. 只口头说明

C. Oral statements only

D. 不保留变更记录

D. No change record maintained

作答 / Answer：_____

第二部分：答案与解析 / Part II: Answers and Explanations

建议完成全部题目后再阅读本部分。 / Complete all questions before reviewing this section.

第一领域：系统与服务器启动（23题）
Domain 1: System and Server Bring-up (23 questions)

1. 答案：B nvidia-smi会查询驱动与GPU管理接口，是安装后的第一线验证工具。
Answer: B nvidia-smi will query the GPU management interface, which is the first-line validation tool once installed.

2. 答案：B BMC提供带外管理，可进行电源控制、硬件监控、远程控制台和事件记录。
Answer: B BMC provides out-of-the-band management that allows power control, hardware monitoring, remote control tables and event recording.

3. 答案：B 带外网络与业务数据平面分离，主机故障时仍可访问BMC或交换机管理接口。
Answer: B The off-site network is separated from the operational data plane and the BMC or switch management interface is still accessible in case of a mainframe failure.

4. 答案：A TPM保护测量值和密钥，可支持平台完整性验证与安全启动。
Answer: A TPM protects measurements and keys that support the verification of the integrity of the platform and its safe start.

5. 答案：ABC 高密度GPU服务器必须同时验证承重、供电、散热气流与接地。
Answer: ABC High-density GPU servers must simultaneously verify loads, power supply, heat flow and location.

6. 答案：B 固件组件彼此有依赖，应依据支持矩阵规划升级顺序并保留回退路径。
Answer: B Solid components are dependent on each other and should be sequenced for upgrades based on the supporting matrix and maintain a regression path.

7. 答案：B 先比较设计容量、冗余要求与实际峰值，避免过载或冗余失效。
Answer: B First, the design capacity, redundancy requirements and actual peaks are compared to avoid overloading or redundancy failures.

8. 答案：B 分离进风与排风可降低回流与热点，提高冷却效率。
Answer: B Separating into wind and wind can reduce backflows and hot spots and increase cooling efficiency.

9. 答案：ABCD 这些项目共同覆盖设备识别、环境、显存可靠性与GPU互连。
Answer: ABCD These projects jointly cover equipment identification, the environment, visible reliability and GPU interconnection.

10. 答案：A 即使驱动尚未正常，lspci仍可用来确认PCIe层是否看见设备。
Answer: A Even if the drive is not normal, lspci can be used to confirm whether the PCIe layer has seen the equipment.

11. 答案：B PCIe都未枚举时，应先查硬件连接、供电和平台资源，而不是上层软件。
Answer: B When PCIe is not listed, hardware connection, power supply and platform resources should be checked first, rather than top-level software.

12. 答案：A lspci -vv可显示LnkCap与LnkSta，用于比较链路能力和实际协商状态。
Answer: A lspci-vvv shows LnkCap and LnkSta for comparison of chain capabilities and actual state of consultation.

13. 答案：ABCD 收发器必须在接口、介质、距离及设备兼容性上同时匹配。
Answer: ABCD The transceiver shall be matched by an interface, medium, distance and equipment compatibility.

14. 答案：B 光学规格不匹配会导致光功率预算或传输模式不符合要求。
Answer: B The failure to match optical specifications may result in a failure of the light budget or mode of transmission.

15. 答案：B 叶脊结构提供一致跳数并适合大规模GPU间东西向通信。
Answer: B The ridge structure provides a consistent number of jumps and is suitable for communication between large-scale GPUs.

16. 答案：B 平面分离有助于性能、可用性和安全管理。
Answer: B Area separation facilitates performance, usability and security management.

17. 答案：ABCD 安全且可审计的BMC基线需要网络、身份、时间、日志及访问限制。
Answer: ABCD A secure and auditable BMC baseline requires network, identity, time, log and access restrictions.

18. 答案：B 重复PSU事件通常从物理供电链与硬件状态开始排查。
Answer: B Repetition of PSU events usually begins with physical power chain and hardware status.

19. 答案：A 高功耗双宽GPU需要正确供电、机械固定和气流空间。
Answer: A High-capacity, double-wide GPS requires proper power supply, mechanical fixation and air current space.

20. 答案：B 存储接入需先确保连通、权限、协议与挂载配置正确。
Answer: B Storage access requires ensuring that access, privileges, protocols and mounted configurations are correct.

21. 答案：ABCD 完整基线便于后续追踪变更、比较性能及快速排错。
Answer: ABCD A complete baseline allows for follow-up of changes, comparison of performance and quick blundering.

22. 答案：B 应以证据确认兼容性或升级失败，并使用预先验证的回退方案。
Answer: B Compatibility or failure to upgrade should be established by evidence, and a pre-certified retreat programme should be used.

23. 答案：A 可重复且记录完整的基线是后续集群安装和故障隔离的依据。
Answer: A Repeatable and fully documented baselines are the basis for subsequent cluster installation and failure isolation.

第二领域：实体层管理（4题）
Domain 2: Physical Layer Management (4 questions)

24. 答案：B MIG可把一张支持的GPU分割成多个隔离的计算与显存实例。
Answer: B MIG can divide a supported GPU into several isolated calculations and visible examples.

25. 答案：ABCD MIG不只是切分，还涉及硬件支持、工作负载尺寸、调度和运维。
Answer: ABCD The MIG is not just about splits, but also about hardware support, load size, movement control and transportation.

26. 答案：A BlueField可在DPU上处理基础设施服务，释放主机CPU并强化隔离。
Answer: A BlueField can handle infrastructure services on the DPU, release the host CPU and enhance isolation.

27. 答案：B 硬件分区变化必须被设备插件或调度器重新发现，资源视图才会一致。
Answer: B The changes in the hardware partition must be rediscovered by the equipment plugin or scheduler in order for the resource view to be consistent.

第三领域：控制平面安装与配置（14题）
Domain 3: Control Plane Installation and Configuration (14 questions)

28. 答案：A Base Command Manager用于集群供应、配置、监控和管理。
Answer: A Base Command Manager is used for cluster supply, configuration, monitoring and management.

29. 答案：A HA确保主要管理节点故障时关键管理能力可继续服务。
Answer: A HA ensures that critical management capacity can continue to be serviced in the event of failure of the main management nodes.

30. 答案：ABCD 批量供应依赖镜像、网络、节点分类及软硬件兼容基线。
Answer: ABCD Batch supply relies on mirrors, networks, nodal classifications and compatible baselines for software and hardware.

31. 答案：A sinfo显示分区、节点数量和idle/down/alloc等状态。
Answer: A Sinfo shows partitions, number of nodes and the status of idle/down/alloc etc.

32. 答案：A squeue用于观察作业队列、状态、用户、分区和运行时间。
Answer: A The squeue is used to observe the operational queue, state, user, partition and running time.

33. 答案：A Enroot提供轻量容器运行能力，Pyxis负责与Slurm集成。
Answer: A Enroot provides lightweight container capacity, and Pyxis is responsible for integration with Slurum.

34. 答案：A 驱动内核模块必须与正在运行的内核兼容并成功加载。
Answer: A The driver's kernel module must be compatible with the active kernel and successfully loaded.

35. 答案：A DOCA是开发和管理BlueField加速网络、存储与安全服务的软件框架。
Answer: A DOCA is the software framework for the development and management of BlueField ' s accelerated network, storage and security services.

36. 答案：A --gpus all把GPU暴露给容器，容器内nvidia-smi可验证运行链路。
Answer: A -gpus all exposed the GPU to the packaging, where nvidia-smi can verify the operational link.

37. 答案：ABCD 工具、认证、连通性及授权缺一不可。
Answer: ABCD Tools, certification, connectivity and delegation of authority are essential.

38. 答案：A Kubernetes必须发现并公布GPU资源，Pod才能请求nvidia.com/gpu。
Answer: A Kubernetes must identify and disclose GPU resources before Pod can request nvidia.com/gpu.

39. 答案：A 容器提供用户态CUDA库，但实际仍使用主机内核驱动。
Answer: A The container provides the user-state CUDA library, but the host's internal nuclear drive is still used.

40. 答案：ABCD 分类用于以可重复方式管理具有相同角色或配置的节点集合。
Answer: ABCD Classifications are used to manage the nodal collections with the same role or configuration in a repeatable manner.

41. 答案：A 端到端作业能同时验证管理、节点、调度、容器和GPU访问链路。
Answer: A End-to-end operations allow simultaneous validation of management, nodes, movement control, containers and GPU access links.

第四领域：集群测试与验证（25题）
Domain 4: Cluster Test and Verification (25 questions)

42. 答案：A HPL通过求解大型稠密线性方程组报告高性能浮点运算结果。
Answer: A HPL reports high performance float results by solving large dense linear equation groups.

43. 答案：A Rmax是实测性能，常与理论Rpeak比较效率。
Answer: A Rmax is a measured performance that is often more efficient than the theory Rpeak.

44. 答案：A 持续高负载会暴露热降频、掉卡、ECC及电源问题。
Answer: A Continued high loads can expose heat and precipitation frequencies, drop cards, ECCs and power problems.

45. 答案：A NCCL Tests用于all-reduce等GPU集合通信性能验证。
Answer: A NCCL Tests are used to validate GPU group communications performances for all-reduce and others.

46. 答案：A 这些因素直接决定GPU间路径与跨节点通信性能。
Answer: A These factors directly determine the routing and cross-node communication performance of the GPU.

47. 答案：A 拓扑矩阵可帮助确认NVLink、PCIe及NUMA亲和关系。
Answer: A An extended matrix can help confirm the relationship between NVLink, PCIe and NUMA.

48. 答案：ABCD 这些指标可判断负载是否充分以及是否出现热、功耗或硬件异常。
Answer: ABCD These indicators determine whether the load is sufficient and whether heat, effort or hardware anomalies occur.

49. 答案：A DCGM提供GPU诊断、健康监控和管理能力。
Answer: A DCGM provides GPU diagnostic, health surveillance and management capabilities.

50. 答案：A ClusterKit帮助自动化节点和集群级健康与性能检查。
Answer: A ClusterKit helps automated nodes and cluster-level health and performance checks.

51. 答案：ABCD 链路、介质、错误和控制服务共同影响fabric健康。
Answer: ABCD Linkage, media, error and control services together affect the health of fabric.

52. 答案：A ibstat显示CA、端口状态、LID、速率等基本信息。
Answer: A Ibstat displays basic information such as CA, port status, LD, speed, etc.

53. 答案：A CRC常指向物理链路、模块、线缆或端口问题。
Answer: A CRC often points to physical links, modules, cables or ports.

54. 答案：A 版本不兼容可能造成链路、协议或管理异常。
Answer: A Incompatibility of the version may result in a chain, agreement or management anomaly.

55. 答案：A DPU验证需要同时覆盖固件、软件、模式、接口和连通性。
Answer: A DPU validation requires simultaneous coverage of solids, software, models, interfaces and connectivity.

56. 答案：ABCD 这些信息可判断模块兼容性及光学链路健康。
Answer: ABCD This information will determine the compatibility of the modules and the health of the optical links.

57. 答案：A 训练通信主要在节点间横向流动，因此称东西向fabric。
Answer: A The training communication is mainly transversal and therefore referred to as fabric.

58. 答案：A 以健康节点为基线比较可快速定位热、硬件或网络差异。
Answer: A Health nodes are used as a baseline against which to quickly locate heat, hardware or network differences.

59. 答案：A NeMo工作负载可验证更贴近实际AI训练的软件和通信路径。
Answer: A NeMo loads validate software and communication paths closer to actual AI training.

60. 答案：ABCD AI数据管线可能同时受带宽、小文件、延迟和并发扩展影响。
Answer: ABCD The AI data pipeline may be affected by bandwidth, small files, delays and simultaneous extension.

61. 答案：A 单客户端结果无法代表集群同时访问时的扩展能力。
Answer: A Single-client results do not represent the ability of clusters to expand upon simultaneous visits.

62. 答案：A 可比较的基准必须控制变量并保留上下文。
Answer: A Comparable benchmarks must control variables and retain context.

63. 答案：ABCD 分层验证才能定位问题并确认端到端生产能力。
Answer: ABCD Layer certification is needed to locate problems and to identify end-to-end production capacity.

64. 答案：A 通信正常后应转向数据管线、主机亲和与应用层效率。
Answer: A When communications are normal, they should be redirected to data pipelines, mainframe connections and application levels.

65. 答案：A 灯亮只能证明有链路，不能证明连到正确端口。
Answer: A The light can only prove a link and not a link to the correct port.

66. 答案：A 预先定义的验收标准能避免主观判断并支持一致交付。
Answer: A Predefined acceptance criteria avoid subjective judgement and support consistent delivery.

第五领域：故障排除与优化（9题）
Domain 5: Troubleshoot and Optimize (9 questions)

67. 答案：A PCIe层未枚举时优先定位硬件、供电或平台问题。
Answer: A Priority positioning hardware, power supply or platform problems when not listed on the PCIe floor.

68. 答案：A Xid代码和上下文是区分软件、链路或硬件问题的重要证据。
Answer: A The Xid code and context are important evidence to distinguish between software, chain or hardware problems.

69. 答案：ABCD 温度、时钟、限制原因和功耗共同解释降频行为。
Answer: ABCD Temperatures, clocks, limiting causes and power consumption together explain low frequency behaviour.

70. 答案：A 正确NUMA亲和可减少跨socket数据移动和延迟。
Answer: A Correct Numa proxies and reduces the movement and delay of data across socket.

71. 答案：A 性能可能卡在客户端参数、元数据、磁盘或条带策略。
Answer: A Performance may be contained in client parameters, metadata, disks or strip policies.

72. 答案：A 应依据热插拔与冗余要求安全更换，并验证告警清除。
Answer: A The requirements should be replaced on the basis of thermal injection and redundancy, and the clearance of the alarm should be verified.

73. 答案：ABCD 更换后必须从识别、版本、健康到互连性能重新验收。
Answer: ABCD The replacement must be based on re-acceptance from identification, version, health and interconnectivity.

74. 答案：A 双路平台常因跨NUMA访问产生瓶颈，亲和配置是通用重点。
Answer: A The two-way platform is often bottlenecked by cross-NUMA visits, with a common focus on kinship and configuration.

75. 答案：A 闭环记录与自动化基线能把一次修复转化为持续预防能力。
Answer: A Closed-ring records and automated baselines translate a restoration into a continuous preventive capability.

重点速记

- HPL看FP64计算与满载稳定性；NCCL看GPU集合通信；DCGM看GPU健康。
- lspci看PCIe是否枚举；nvidia-smi看驱动与GPU运行状态；nvidia-smi topo -m看拓扑。
- PCIe都看不到设备时先查物理层、供电、插槽和BIOS资源，不要先查应用。
- 容器使用主机驱动；NVIDIA Container Toolkit负责把GPU能力提供给容器。
- 集群验收应从单节点、节点内互连、跨节点网络、存储，一直验证到代表性AI负载。

官方参考

NVIDIA NCP-AII certification page:
https://www.nvidia.com/en-us/learn/certification/ai-infrastructure-professional/